Name:________________________ HR:____________

6th Grade Math Unit 7: Equations STAR Cards

7

QUESTION	ANSWER
7a) ☆☆☆ 1. What does it mean to *solve* an equation? 2. Solve this equation: $x + 5 = 15$	7a) 1. *Solving* an equation means **finding the value of the variable** that will make the left side equal the right side. 2. Solve: $x + 5 = 15$ *10 + 5 = 15 so* ***x = 10***
7b) ☆☆☆ How can you figure out which of these numbers {2, 5, 6} makes the following equation true? $3x + 2 = 17$	7b) 1st) Substitute each of the numbers for x 2nd) Simplify until you find the number that makes the equation true $3(5) + 2 = 17$ 15 + 2 = 17 17 = 17 ✔ **so x = 5**
7c) ☆☆☆ What is the relationship between an independent and a dependent variable?	7c) The <u>independent variable</u> *causes* the <u>dependent variable</u>

7d)

What are 3 main ways to <u>represent a relationship</u> in algebra?

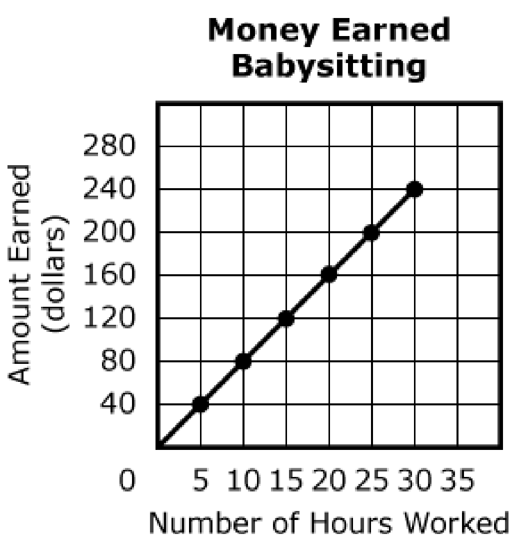
7d)

<u>G</u>raph, <u>E</u>quation, <u>T</u>able

7e)

1. What does the point (20, 160) mean on the graph below?

2. What does the point (5,40) mean on the graph below?



7e)

1. (20,160) means **for every 20 hours you babysit you earn $160.**

2. (5,40) means **for every 5 hours you babysit you earn $40**

7f) ☆☆☆

1. This table is proportional. Find the unit rate.
2. Show how to use the unit rate to find the missing values in the table.

Hours worked (x)	Dollars Earned (y)
2	**10**
6	
	40

7f)

1. 5 dollars for every 1 hour
2.

Hours worked (x)	Dollars Earned (y)
2	10
6 × unit rate =	
	40

÷ unit rate = x

7g) ☆☆☆

If you are given a proportional table, how do you write an equation?

For example:

Hours worked (x)	Dollars Earned (y)
2	**10**
6	
	40

7g)

To write an equation from a proportional table:

1st) Find the $unit\ rate$

2nd) Write: $y = unit\ rate \cdot x$

Example: $y = 5x$